

Summer Bridge Practice Packet

Lock in your 6th-grade math skills and get a head start on 7th. A few clean problems a day all summer keeps your brain sharp and makes the first weeks of Grade 7 feel easy.

Name:

Teacher:

Start date:

Finish date:

What this packet is. This is a summer review-and-preview packet. Parts 1–7 review the most important math you learned in Grade 6 — ratios, fractions, percents, expressions and equations, rational numbers, geometry, and statistics. Parts 8–10 gently *preview* three big ideas you will see early in Grade 7 — integer operations, proportional relationships, and two-step equations with percent change. Each preview part teaches first, then asks you to try.

How to use this packet

- **Pace:** Plan for about **4–6 weeks**. Do roughly **one Part every 2–3 days**, or **6–10 problems a day**. Short and steady beats one giant cram session.
- **Watch one first.** Read the *Worked Example* at the top of each Part before you try the practice.

- **Pick your level.** Every Part has a **LEVEL 1** set (extra support, friendlier numbers) and a **LEVEL 2** set (stretch yourself). Do the main set, then choose.
- **Check yourself.** Finish each Part with the *Quick Check*, then use the *Answer Keys* at the back to grade your work.
- **Stuck?** Tap the blue *reference chips* in each Part header to reopen the matching lesson notes on the Neft Teacher site.
- **Check off your Parts:** ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10
☐ Spiral A ☐ Spiral B

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10. Part 10 — PREVIEW: Two-Step Equations & Percent Change · *7.EE, 7.RP*
11. Cumulative / Spiral Review (Sets A & B)
12. Answer Keys (all Parts + Spiral)

Standards & Skills Map

Part	Skill focus	Standards	Where to review
1	Ratio language, tables & tape diagrams, unit rate, equivalent ratios	6.RP.A	Unit 3 Notes
2	Four operations with fractions & mixed numbers, dividing fractions, decimal operations	6.NS.A, 6.NS.B	Unit 2 Notes
3	Percent of a number, find the whole, FDP conversions, percent word problems	6.RP.A.3	Unit 4 Notes
4	Evaluate/write expressions, properties, one-step equations, one-step inequalities	6.EE	Unit 6, Unit 7
5	Integers, opposites, absolute value, ordering, four-quadrant plotting, distance	6.NS.C	Unit 9 Notes

Part	Skill focus	Standards	Where to review
6	Area of polygons & composites, nets & surface area, volume with fractional edges	6.G	Unit 5, Unit 10
7	Statistical questions, center & spread, MAD, dot plots, histograms, box plots	6.SP	Unit 8 Notes
8	PREVIEW — add, subtract, multiply, divide signed numbers	7.NS	Unit 9, Vocabulary
9	PREVIEW — constant of proportionality, $y = kx$, proportional vs. not	7.RP	Unit 3, Unit 4
10	PREVIEW — two-step equations; markup, discount, tax, tip, percent change	7.EE, 7.RP	Unit 7, Unit 4

Where to review — how the reference chips work

Each Part header has a row of blue **reference chips**. These are direct links to the matching notes, slides, and practice on the Neft Teacher site. If a skill feels rusty, tap a chip *before* you practice to relearn it. *Vocabulary* opens the math vocab deck, and *Spiral Review* opens extra mixed practice. (Links work on the web version; on a printed copy, type the address shown.)

1

Ratios, Rates & Unit Rate

6.RP.A

REVIEW:

Unit 3 Notes (Ratios)

Unit 4 Notes (Rates)

Vocabulary

VOCABULARY

Ratio	A comparison of two quantities, like 3 cats to 4 dogs (written 3:4 or $\frac{3}{4}$).
Rate	A ratio that compares two <i>different</i> units, like miles per hour.
Unit rate	A rate with a denominator of 1, like 60 miles in 1 hour.
Equivalent ratios	Ratios that show the same comparison, like 2:3 and 4:6.
Tape diagram	A bar model that shows the parts of a ratio as equal-size boxes.
Ratio table	A table of equivalent ratios you build by multiplying or dividing both numbers.

WATCH ONE FIRST

Example: A smoothie uses 3 cups of berries for every 2 cups of yogurt. How much yogurt is needed for 12 cups of berries, and what is the unit rate of yogurt per cup of berries?

Write the ratio: berries : yogurt = 3 : 2.

Scale up: $3 \times 4 = 12$ berries, so $2 \times 4 = 8$ cups of yogurt.

Unit rate of yogurt per berry = $2 \div 3 = \frac{2}{3}$ cup of yogurt for each cup of berries.

Answer: 8 cups of yogurt; unit rate = $\frac{2}{3}$ cup yogurt per cup of berries.

Set A • Ratio language & equivalent ratios

Write each ratio, then complete.

- Write the ratio of vowels to consonants in the word "MATH."
- A bag has 5 red and 8 blue marbles. Write the ratio red : blue.
- Write the ratio blue : total marbles from problem 2.
- Fill in: $2 : 5 = 8 : \underline{\hspace{1cm}}$
- Fill in: $6 : 9 = \underline{\hspace{1cm}} : 3$
- Fill in: $4 : 7 = 12 : \underline{\hspace{1cm}}$
- Are 3 : 4 and 9 : 16 equivalent? Yes or no — explain in a few words.
- Write two ratios equivalent to 5 : 2.

Set B · Ratio tables & tape diagrams

Build the table or read the model.

1. Complete the table for $2 : 3$ — (2,3), (4,___), (6,___), (___,12).
2. Complete the table for $5 : 4$ — (5,4), (10,___), (___,12), (20,___).
3. A tape diagram shows 4 boxes of apples to 1 box of oranges. If each box holds 6 fruit, how many apples?
4. A recipe uses flour : sugar = $3 : 1$. For 9 cups of flour, how much sugar?
5. Paint mixes red : white = $2 : 5$. For 10 cups of red, how much white?
6. The ratio of boys to girls is $3 : 5$. If there are 24 students total, how many girls? (Hint: 8 equal parts.)
7. A class has cats : dogs = $7 : 3$. If there are 21 cats, how many dogs?

Set C · Unit rate & rates

Find the "per 1" amount.

1. A car travels 150 miles in 3 hours. Find the unit rate (miles per hour).
2. 12 apples cost \$6. Find the cost per apple.
3. A printer prints 90 pages in 6 minutes. Pages per minute?
4. \$45 for 5 movie tickets. Price per ticket?
5. A runner goes 8 km in 50 minutes. Minutes per km?
6. Which is the better buy: 20 oz for \$4.00 or 30 oz for \$5.40? Show the unit prices.

Set D · Find and fix

Spot the mistake.

1. A student says $3 : 4 = 6 : 7$ because they "added 3 and 3, then 3 and 3." Explain the error and give the correct second ratio.
2. A student says 100 miles in 4 hours is a unit rate of 4 miles per hour. Find and fix the mistake.

Level 1 · Scaffolded

LEVEL 1

1. Write the ratio of 2 stars to 6 moons in lowest terms.
2. Fill in: $1 : 3 = 4 : \underline{\hspace{1cm}}$
3. 6 cookies cost \$3. Cost per cookie?
4. A table shows $1 : 2$. Complete: (1,2), (2,___), (3,___).

Level 2 · Stretch

LEVEL 2

1. Smoothie A is 2 cups fruit to 3 cups juice; Smoothie B is 3 cups fruit to 5 cups juice. Which is fruitier? Use unit rates to justify.
2. A jar of nickels and dimes has nickels : dimes = $3 : 2$ and is worth \$1.40 in all. How many nickels and how many dimes? (Hint: try a ratio table.)

QUICK CHECK

1. Fill in: $4 : 6 = \underline{\hspace{1cm}} : 18$
2. 240 miles in 4 hours — miles per hour?
3. boys : girls = $2 : 3$, total 30 students — how many boys?
4. Better buy: 16 oz for \$3.20 or 24 oz for \$4.32?

REVIEW:

Unit 2 Notes (Fraction Division)

Unit 1 Notes (Number Sense)

Spiral Review

VOCABULARY

Reciprocal

The "flip" of a fraction; $\frac{3}{4}$ and $\frac{4}{3}$ are reciprocals (they multiply to 1).

Mixed number

A whole number and a fraction together, like $2 \frac{1}{2}$.

Improper fraction

A fraction whose top is greater than or equal to its bottom, like $\frac{7}{2}$.

Common denominator

The same bottom number used to add or subtract fractions.

Simplify

To write a fraction in lowest terms by dividing top and bottom by a common factor.

Quotient

The answer to a division problem.

WATCH ONE FIRST

Example 1 (divide fractions): $\frac{3}{4} \div \frac{2}{3}$.

Keep the first fraction: $\frac{3}{4}$.

Change $\div$ to $\times$ and flip the second to its reciprocal: $\frac{3}{4} \times \frac{3}{2}$.

Multiply across: $(3 \times 3) / (4 \times 2) = \frac{9}{8} = 1 \frac{1}{8}$.

Example 2 (decimals): $4.6 \times 0.3 = 1.38$ (multiply $46 \times 3 = 138$, then place 2 decimal digits).

Set A · Add & subtract fractions Common denominator; simplify.

1. $\frac{1}{2} + \frac{1}{3}$

3. $\frac{5}{6} - \frac{1}{3}$

2. $\frac{3}{4} + \frac{1}{6}$

4. $\frac{7}{8} - \frac{1}{2}$

5. $2\frac{1}{4} + 1\frac{1}{2}$

7. $3 - \frac{2}{3}$

6. $4\frac{1}{3} - 1\frac{5}{6}$

Set B · Multiply fractions Multiply across; simplify.

1. $\frac{2}{3} \times \frac{3}{5}$

4. $1\frac{1}{2} \times \frac{2}{3}$

2. $\frac{4}{9} \times \frac{3}{8}$

5. $2\frac{1}{4} \times 1\frac{1}{3}$

3. $\frac{5}{6} \times \frac{2}{5}$

6. $\frac{2}{7} \times 14$

Set C · Divide fractions (keep–change–flip) Show your reciprocal step.

1. $\frac{1}{2} \div \frac{1}{4}$

5. $2\frac{1}{2} \div \frac{1}{2}$

2. $\frac{3}{5} \div \frac{2}{3}$

6. $3\frac{3}{4} \div 1\frac{1}{4}$

3. $\frac{4}{9} \div \frac{2}{3}$

7. $6 \div \frac{2}{3}$

4. $\frac{5}{8} \div \frac{1}{4}$

8. How many $\frac{3}{4}$ -cup servings are in 6 cups of rice?

Set D · Decimal operations Line up or place decimals carefully.

1. $3.4 + 12.75$

4. 0.25×0.6

2. $20 - 6.38$

5. $9.6 \div 0.3$

3. 5.6×0.4

6. $14.4 \div 1.2$

Set E · Word problems & find-and-fix Read carefully.

1. A ribbon is $7\frac{1}{2}$ ft long. You cut pieces that are $\frac{3}{4}$ ft each. How many pieces?

2. A recipe needs $\frac{2}{3}$ cup of oil. You triple it. How much oil total?

3. Find and fix: a student says $\frac{1}{2} \div \frac{1}{4} = \frac{1}{8}$ because "you multiply the fractions." What did they forget?

Level 1 · Scaffolded LEVEL 1

1. $\frac{1}{4} + \frac{1}{4}$

3. $\frac{1}{2} \div \frac{1}{4}$ (flip the second!)

2. $\frac{1}{2} \times \frac{1}{2}$

4. $2.5 + 1.5$

Level 2 · Stretch LEVEL 2

1. $(\frac{2}{3} + \frac{1}{4}) \div \frac{5}{6}$

2. A trail is $4\frac{1}{2}$ miles. You hike $\frac{2}{3}$ of it before lunch. How many miles are left after lunch?

QUICK CHECK

1. $\frac{5}{6} - \frac{1}{4}$

3. $2\frac{1}{2} \div \frac{3}{4}$

2. $\frac{3}{4} \times \frac{8}{9}$

4. $7.2 \div 0.9$

REVIEW:

Unit 4 Notes (Percents & Rates)

Unit 3 Notes (Ratios)

Vocabulary

VOCABULARY

Percent

A ratio out of 100. 25% means 25 per 100, or 25/100.

Part

The piece you are finding or comparing (the result of the percent).

Whole

The full 100% amount you take the percent of.

Convert

To rewrite a number as a fraction, decimal, or percent.

Percent of a number

Multiply: percent (as a decimal) $\times$ the number.

Benchmark percent

An easy percent like 10%, 25%, 50% you can use to estimate.

WATCH ONE FIRST

Example 1 (percent of a number): What is 30% of 80?

Write 30% as a decimal: 0.30.

Multiply: $0.30 \times 80 = 24$.

Example 2 (find the whole): 15 is 25% of what number?

$25\% = 0.25$, so $0.25 \times \text{whole} = 15$.

Whole = $15 \div 0.25 = 60$.

Set A · Fraction–decimal–percent Convert each.

1. Write $\frac{1}{2}$ as a decimal and a percent.
2. Write $\frac{3}{4}$ as a decimal and a percent.
3. Write 0.6 as a fraction and a percent.

4. Write 45% as a decimal and a fraction (lowest terms).
5. Write $\frac{1}{5}$ as a percent.
6. Write 0.08 as a percent.
7. Write $\frac{7}{10}$ as a percent.
8. Write 12.5% as a fraction (lowest terms).

Set B · Percent of a number Decimal $\times$ number.

1. 20% of 50
2. 10% of 90
3. 75% of 40
4. 5% of 200
5. 60% of 35
6. 150% of 20
7. 15% of 80
8. 8% of 250

Set C · Find the whole Part $\div$ percent.

1. 12 is 25% of what number?
2. 30 is 50% of what number?
3. 9 is 10% of what number?
4. 18 is 75% of what number?
5. 40 is 20% of what number?
6. 7 is 5% of what number?

Set D · Percent word problems Real situations.

1. A jacket costs \$60. It is 25% off. How much is taken off, and what is the sale price?
2. A team won 18 of 24 games. What percent did they win?
3. A class has 30 students; 40% walk to school. How many walk?
4. A tip is 20% of a \$35 bill. How much is the tip?
5. A \$2.00 bag of chips goes up 15%. What is the new price?

Level 1 · Scaffolded **LEVEL 1**

1. 50% of 20
2. 10% of 60
3. Write $\frac{1}{4}$ as a percent.
4. 25% of 40

Level 2 · Stretch **LEVEL 2**

1. A phone is \$200. With a 30% discount and then 6% sales tax on the sale price, what is the final cost?
2. 32 is 80% of a number. The number is what percent of 100? (Find the whole first.)

QUICK CHECK

1. 15% of 60
2. Write 0.35 as a percent.
3. 21 is 70% of what number?
4. A \$50 game is 20% off. Sale price?

REVIEW:

Unit 6 Notes (Expressions)

Unit 7 Notes (Equations & Inequalities)

Spiral Review

VOCABULARY

Expression

A math phrase with numbers, variables, and operations (no equals sign).

Variable

A letter that stands for an unknown number, like x .

Coefficient

The number multiplied by a variable; in $5x$ the coefficient is 5.

Evaluate

To find the value of an expression by plugging in a number.

Equation

A statement that two expressions are equal (has an $=$ sign).

Inequality

A statement using $<$, $>$, $\leq$, or $\geq$ to compare.

WATCH ONE FIRST

Example 1 (evaluate): Evaluate $3x + 5$ when $x = 4$.

Substitute: $3(4) + 5$.

Multiply, then add: $12 + 5 = 17$.

Example 2 (solve one-step): $x + 7 = 12$.

Undo $+7$ by subtracting 7 from both sides: $x = 12 - 7 = 5$.

Set A · Write & read expressions Translate words to math.

- Write: "7 more than a number n ."
- Write: "the product of 4 and x ."
- Write: "a number y divided by 5."
- Write: "6 less than twice a number m ."
- Identify the coefficient and constant in $9x + 2$.

6. How many terms are in $3a + 4b + 7$?

Set B · Evaluate expressions

Use order of operations.

1. $2x + 3$ when $x = 5$

4. $5(a + 2)$ when $a = 4$

2. $4n - 1$ when $n = 6$

5. $30 \div m$ when $m = 6$

3. $x^2 + 1$ when $x = 3$

6. $3x + 2y$ when $x = 2, y = 5$

Set C · Properties & equivalent expressions

Use the distributive property and combine like terms.

1. Use the distributive property: $3(x + 4)$.

4. Combine like terms: $7y + 2 + y$.

2. Use the distributive property: $5(2a + 1)$.

5. Factor out the GCF: $6x + 9$.

3. Combine like terms: $4x + 3x$.

6. Are $2(x + 3)$ and $2x + 6$ equivalent? Explain.

Set D · Solve one-step equations

Undo the operation.

1. $x + 8 = 15$

4. $x/3 = 7$

2. $x - 4 = 10$

5. $x + 2.5 = 9$

3. $6x = 42$

6. $5x = 20$

Set E · Write & graph one-step inequalities

Describe the solution.

1. Write an inequality: "a number x is at least 10."

2. Write an inequality: "you must be under 13 (age a) to enter."

3. Solve and describe the solution: $x + 5 > 12$.

4. Solve and describe the solution: $x - 3 \leq 4$.

5. Solve: $4x \leq 20$. Then name two values of x that work.

Level 1 · Scaffolded

LEVEL 1

1. Evaluate $x + 3$ when $x = 5$.

3. Solve $3x = 12$.

2. Solve $x + 2 = 9$.

4. Write: "5 more than n ."

Level 2 · Stretch

LEVEL 2

1. Simplify, then evaluate when $x = 2$: $3(x + 4) + 2x$.
2. Wendy has \$50 and saves \$8 per week. Write an expression for her total after w weeks, then find the total after 6 weeks.

QUICK CHECK

1. Evaluate $2x + 7$ when $x = 6$.
2. Solve $x/4 = 5$.
3. Distribute: $4(x + 3)$.
4. Solve and describe: $x + 6 \geq 10$.

REVIEW:

Unit 9 Notes (Rational Numbers)

Vocabulary

VOCABULARY

Integer

A whole number or its opposite: ... -2, -1, 0, 1, 2 ...

Opposite

The number the same distance from 0 but on the other side; opposite of -3 is 3.

Absolute value

A number's distance from 0, always 0 or positive. $|-4| = 4$.

Coordinate plane

A grid made by a horizontal x-axis and vertical y-axis.

Ordered pair

(x, y) tells you how far across, then up or down.

Quadrant

One of the four regions of the plane, numbered I–IV.

WATCH ONE FIRST

Example 1 (order & absolute value): Order from least to greatest: -3, 2, -5, 0. Then find $|-5|$.

On a number line, more-negative is smaller: $-5 < -3 < 0 < 2$.

$|-5| = 5$ (distance from 0).

Example 2 (plot & distance): Distance between (2, 3) and (2, -4).

Same x, so count vertically: from 3 down to -4 is $3 + 4 = 7$ units.

Set A · Integers, opposites & absolute value Use the number line.

1. What is the opposite of 7?

2. What is the opposite of -12?

- Find $|-9|$.
- Find $|15|$.
- Which is greater: -8 or -3 ?
- Which is greater: $|-6|$ or $|4|$?
- A temperature of -5°F rises 8 degrees. What is the new temperature?
- Write the integer for "12 feet below sea level."

Set B · Order rational numbers Least to greatest unless told otherwise.

- Order: $-2, 5, -7, 1$.
- Order: $0, -1, -4, 3$.
- Order: $1/2, -1/2, 0, 1$.
- Order: $-1.5, -1, -2, 0$.
- Compare with $<$, $>$, or $=$: -3 ____ -6 .
- Compare: $2/3$ ____ 0.6 .

Set C · Plot in four quadrants Name the quadrant.

- Which quadrant is $(3, -2)$ in?
- Which quadrant is $(-4, -1)$ in?
- Which quadrant is $(-2, 5)$ in?
- Where is $(0, -3)$? (On which axis?)
- Reflect $(3, 4)$ across the x-axis. New point?
- Reflect $(-2, 5)$ across the y-axis. New point?

Set D · Distance on the plane & find-and-fix Same x or same y → count or add.

- Distance between $(1, 2)$ and $(1, 9)$.
- Distance between $(-3, 4)$ and $(5, 4)$.
- Distance between $(-2, -1)$ and $(-2, 6)$.
- Find and fix: a student says $|-8| = -8$. What is the correct value and why?

Level 1 · Scaffolded **LEVEL 1**

- Opposite of 4?
- Find $|-3|$.
- Which is greater: -2 or 1 ?
- Which quadrant is $(2, 3)$ in?

Level 2 · Stretch **LEVEL 2**

- A diver is at -30 ft and rises 12 ft, then drops 7 ft. What is the diver's new depth?
- A rectangle has corners $(-3, 2)$, $(4, 2)$, $(4, -1)$, $(-3, -1)$. Find its length, width, and area.

QUICK CHECK

1. Find $|-11|$.
2. Order least to greatest: $-4, 2, -9, 0$.
3. Which quadrant is $(-5, -2)$ in?
4. Distance between $(2, -3)$ and $(2, 5)$.

REVIEW:

Unit 5 Notes (Area)

Unit 10 Notes (Volume & Surface Area)

VOCABULARY

Area

The number of square units that cover a flat shape.

Base & height

A side and the perpendicular distance to the opposite side or vertex.

Composite figure

A shape made of two or more simpler shapes.

Net

A flat pattern that folds up into a 3-D solid.

Surface area

The total area of all the faces of a 3-D solid.

Volume

The amount of space inside a solid, in cubic units.

WATCH ONE FIRST

Example 1 (triangle area): A triangle has base 10 cm, height 6 cm.

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2.$$

Example 2 (volume, fractional edges): A box is $2\frac{1}{2}$ by 3 by 4 units.

$$\text{Volume} = l \times w \times h = 2.5 \times 3 \times 4 = 30 \text{ cubic units.}$$

Set A · Area of polygons Use the right formula.

1. Rectangle: 8 by 5. Area?
2. Triangle: base 12, height 7. Area?
3. Parallelogram: base 9, height 4. Area?
4. Triangle: base 5, height 6. Area?
5. Trapezoid: bases 6 and 10, height 4.
 $\text{Area} = \frac{1}{2}(b_1 + b_2)h.$
6. Parallelogram: base 11, height 8. Area?

Set B · Composite figures

Split into pieces, then add.

1. An L-shape is a 10 by 6 rectangle with a 4 by 3 rectangle removed from one corner. Find its area.
2. A figure is a 8 by 4 rectangle with a triangle (base 8, height 3) on top. Find the total area.
3. A garden is a 12 by 5 rectangle plus a half-circle-free square 5 by 5 added to one end. Find the total area (two rectangles).

Set C · Nets & surface area

Add up all faces.

1. A cube has edge 3 cm. Surface area = $6 \times (\text{edge})^2$. Find it.
2. A rectangular prism is 4 by 3 by 2. Surface area = $2(lw + lh + wh)$. Find it.
3. A cube has edge 5 in. Find its surface area.
4. A net shows 6 squares each 2 by 2. What solid is it, and what is its surface area?

Set D · Volume (incl. fractional edges)

$l \times w \times h$.

1. Prism: 5 by 4 by 3. Volume?
2. Cube: edge 4. Volume?
3. Prism: $1\frac{1}{2}$ by 3 by 2. Volume?
4. Prism: $2\frac{1}{2}$ by 2 by 4. Volume?
5. Prism: $1\frac{1}{2}$ by $1\frac{1}{2}$ by $1\frac{1}{2}$. Volume?
6. A box holds unit cubes of edge $1\frac{1}{2}$. How many $1\frac{1}{2}$ -cubes fit in a 1 by 1 by 1 cube?

Level 1 · Scaffolded

LEVEL 1

1. Rectangle 6 by 4. Area?
2. Triangle base 8, height 4. Area? (use $\frac{1}{2}bh$)
3. Cube edge 2. Volume?
4. Cube edge 2. Surface area? ($6 \times \text{side}^2$)

Level 2 · Stretch

LEVEL 2

1. A fish tank is 20 by 10 by 12 in. How many cubic inches of water fill it to the top? If $231 \text{ in}^3 = 1 \text{ gallon}$, about how many gallons (round to the nearest whole)?
2. A trapezoid has bases 8 and 14 and height 6. A second identical trapezoid is placed beside it. What is the combined area?

QUICK CHECK

1. Triangle base 10, height 9. Area?
2. Trapezoid bases 4 and 8, height 5. Area?
3. Prism 3 by 3 by 6. Volume?
4. Cube edge 4. Surface area?

REVIEW:

Unit 8 Notes (Statistics & Data)

Spiral Review

VOCABULARY

Statistical question

A question that expects a variety of answers, like "How tall are students?"

Mean

The average: add the values, divide by how many.

Median

The middle value when the data is in order.

Mode

The value that appears most often.

Range

The biggest value minus the smallest value.

MAD

Mean absolute deviation: the average distance of values from the mean.

WATCH ONE FIRST

Example 1 (center & spread): Data: 4, 8, 6, 8, 4.

Order it: 4, 4, 6, 8, 8. Median = middle = 6. Mode = 4 and 8. Range = $8 - 4 = 4$.

Mean = $(4+8+6+8+4) \div 5 = 30 \div 5 = 6$.

Example 2 (MAD): For 4, 4, 6, 8, 8 with mean 6: distances are 2, 2, 0, 2, 2. MAD = $(2+2+0+2+2) \div 5 = 8 \div 5 = 1.6$.

Set A · Statistical questions Statistical or not?

1. Is "How old am I?" a statistical question? Explain.
2. Is "How many minutes do 6th graders sleep?" statistical?
3. Write one statistical question about pets.
4. Is "What is the high temperature today?" statistical?

Set B · Mean, median, mode, range Order first.

1. Data: 3, 7, 7, 9, 9. Find mean.
2. Same data: find median.
3. Same data: find mode.
4. Same data: find range.
5. Data: 10, 12, 14, 16. Find mean.
6. Data: 5, 5, 8, 12, 20. Find median.
7. Data: 2, 4, 4, 4, 6, 10. Find mode and range.
8. Data: 6, 8, 10, 12. Find mean and median.

Set C · MAD & spread Find the mean first, then average distances.

1. Data: 2, 4, 6, 8. Find the mean, then the MAD.
2. Data: 10, 10, 12, 12. Find the mean, then the MAD.
3. Data set X has range 4; set Y has range 12. Which is more spread out?

Set D · Read displays & describe shape Dot plots, histograms, box plots.

1. A dot plot has dots at 3 (x2), 4 (x4), 5 (x1). How many data points total, and what is the mode?
2. A histogram groups test scores into 60–69, 70–79, 80–89, 90–99 with 3, 8, 10, 4 students. How many students total? Which interval has the most?
3. A box plot has min 2, Q1 5, median 8, Q3 11, max 16. Find the range and the interquartile range (IQR = $Q3 - Q1$).
4. Is a data set "skewed right," "skewed left," or "symmetric" if most values bunch on the low end with a long tail to the high end?

Level 1 · Scaffolded **LEVEL 1**

1. Data: 2, 4, 6. Find the mean.
2. Data: 1, 3, 5, 7, 9. Find the median.
3. Data: 4, 4, 7. Find the mode.
4. Data: 3, 10. Find the range.

Level 2 · Stretch **LEVEL 2**

1. The mean of 5 numbers is 12. Four of them are 10, 14, 9, 15. Find the fifth number.
2. Data: 3, 5, 5, 7, 10. Find the mean and the MAD.

QUICK CHECK

1. Data: 4, 6, 6, 8, 11. Find the mean.
2. Same data: find the median.
3. Same data: find the range.
4. Data: 1, 3, 5, 7. Find the MAD.

GET AHEAD:

Unit 9 Notes (Rational Numbers)

Vocabulary

New in Grade 7 — teach yourself first

In Grade 6 you compared and ordered negative numbers. In Grade 7 you will *add*, *subtract*, *multiply*, and *divide* with them. Read both worked examples, then try the practice. Take it slow — the rules are short, but the signs matter.

VOCABULARY**Signed number**

A positive or negative number.

Sum

The result of adding.

Difference

The result of subtracting.

Additive inverseThe opposite of a number; 5 and -5 add to 0.**Same sign**

Both positive or both negative.

Different signs

One positive and one negative.

WATCH ONE FIRST — ADD & SUBTRACT

Rule (add): Same signs → add and keep the sign. Different signs → subtract and take the sign of the bigger-distance number.

| $-3 + (-5)$: same sign, add $3 + 5 = 8$, keep negative → -8 .

| $-7 + 2$: different signs, $7 - 2 = 5$, the 7 is bigger and negative → -5 .

Rule (subtract): "Keep–Change–Change" — subtracting is adding the opposite.

| $4 - 9 = 4 + (-9) = -5$.

| $-2 - 6 = -2 + (-6) = -8$.

| $3 - (-5) = 3 + 5 = 8$.

WATCH ONE MORE — MULTIPLY & DIVIDE

Rule: Same signs → positive answer. Different signs → negative answer.

| $(-4)(-3) = 12$ (same signs → positive).

| $(-6)(2) = -12$ (different signs → negative).

| $-20 \div 5 = -4$ (different signs → negative).

| $-18 \div (-3) = 6$ (same signs → positive).

Set A · Add integers

Same sign add, different sign subtract.

1. $-4 + (-6)$

5. $-10 + 10$

2. $-8 + 3$

6. $6 + (-9)$

3. $9 + (-5)$

7. $-5 + 12$

4. $-2 + (-7)$

8. $-3 + (-3)$

Set B · Subtract integers

Add the opposite.

1. $3 - 8$

5. $0 - 9$

2. $-2 - 5$

6. $-5 - (-5)$

3. $4 - (-6)$

7. $10 - (-4)$

4. $-7 - (-3)$

8. $-6 - 6$

Set C · Multiply & divide integers Same sign $\rightarrow +$, different $\rightarrow -$.

1. $(-5)(3)$
2. $(-4)(-6)$
3. $(7)(-2)$
4. $-24 \div 6$
5. $-30 \div (-5)$
6. $(-8)(0)$
7. $36 \div (-9)$
8. $(-2)(-2)(-2)$

Set D · Word problems & find-and-fix Negatives are real.

1. The temperature is -3°F at dawn and drops 7 more degrees. What is the new temperature?
2. A submarine is at -120 ft and rises 45 ft. What is its new depth?
3. You owe \$15 (that is -15) and pay back \$9. What is your balance now?
4. Find and fix: a student says $-5 - 3 = -2$. What is the correct answer and why?

Level 1 · Scaffolded LEVEL 1

1. $-2 + (-3)$
2. $5 + (-1)$
3. $2 - 5$
4. $(-3)(2)$

Level 2 · Stretch LEVEL 2

1. Evaluate: $-4 + 7 - (-2) - 9$.
2. Evaluate: $(-3)(4) + (-20) \div (-5)$.

QUICK CHECK

1. $-6 + 10$
2. $-4 - (-9)$
3. $(-7)(-3)$
4. $-28 \div 4$

GET AHEAD:

Unit 3 Notes (Ratios)

Unit 4 Notes (Rates)

New in Grade 7 — teach yourself first

In Grade 6 you found unit rates. In Grade 7 you will use the *constant of proportionality* (called **k**) to write equations like **$y = kx$** and to decide whether two quantities are proportional. Read the examples, then try.

VOCABULARY**Proportional**

Two quantities are proportional if one is always the same number times the other.

Constant of proportionality (k)

The fixed multiplier: $k = y \div x$.

Equation $y = kx$

The rule that connects x and y in a proportional relationship.

Origin

The point $(0, 0)$; a proportional graph is a straight line through it.

Unit rate

The same idea as k : the value of y when $x = 1$.

Table check

Divide y by x in each row; if k is the same, it is proportional.

WATCH ONE FIRST — FIND k AND WRITE $y = kx$

Example: A table shows $x = 2, y = 6$ and $x = 5, y = 15$. Find k and the equation.

$k = y \div x$. Row 1: $6 \div 2 = 3$. Row 2: $15 \div 5 = 3$. Same k !

Since $k = 3$, the equation is $y = 3x$.

Predict: when $x = 8, y = 3(8) = 24$.

WATCH ONE MORE — PROPORTIONAL OR NOT?

Example: Is this table proportional? (1, 4), (2, 8), (3, 10).

$$4 \div 1 = 4, \quad 8 \div 2 = 4, \quad 10 \div 3 \approx 3.33.$$

The ratios are not all equal, so it is **not** proportional.

Set A · Find the constant of proportionality $k = y \div x$.

1. $x = 3, y = 12$. Find k .
2. $x = 4, y = 10$. Find k .
3. $x = 5, y = 20$. Find k .
4. $x = 2, y = 9$. Find k .
5. 6 apples cost \$3. Find k (cost per apple).
6. A car goes 120 miles on 4 gallons. Find k (miles per gallon).

Set B · Write & use $y = kx$ Then predict.

1. If $k = 5$, write the equation.
2. For $y = 4x$, find y when $x = 7$.
3. For $y = 2.5x$, find y when $x = 6$.
4. A table has (1, 6), (2, 12), (3, 18). Write $y = kx$.
5. Using $y = 6x$, find y when $x = 10$.
6. For $y = 3x$, find x when $y = 21$.

Set C · Proportional or not? Check every ratio.

1. Table: (2, 6), (4, 12), (6, 18). Proportional? Find k if so.
2. Table: (1, 3), (2, 5), (3, 7). Proportional? Explain.
3. Table: (3, 15), (5, 25), (8, 40). Proportional? Find k if so.
4. A graph is a straight line that passes through (0, 2), not the origin. Proportional? Explain.

Set D · Word problems Use k .

1. A printer prints 8 pages per minute. Write $y = kx$ for pages y in x minutes, then find pages in 15 minutes.
2. Bananas cost \$0.40 each. Write the equation for cost y of x bananas, then find the cost of 12.

Level 1 · Scaffolded **LEVEL 1**

1. $x = 2, y = 8$. Find k .
2. For $y = 5x$, find y when $x = 3$.

3. If $k = 2$, write the equation.

4. $x = 1$, $y = 7$. Find k .

Level 2 · Stretch

LEVEL 2

1. One table is $y = 4x$; another is $y = 7x$. For $x = 5$, which gives the larger y , and by how much?

2. A recipe is proportional: 3 cups flour make 12 muffins. Write $y = kx$ (muffins per cup), then find cups needed for 40 muffins.

QUICK CHECK

1. $x = 4$, $y = 28$. Find k .

3. Is $(2, 6)$, $(4, 12)$, $(5, 14)$ proportional?

2. For $y = 6x$, find y when $x = 9$.

4. Write $y = kx$ for $(1, 9)$, $(2, 18)$.

PREVIEW: Two-Step Equations & Percent Change

7.EE · 7.RP

GET AHEAD:

Unit 7 Notes (Equations)

Unit 4 Notes (Percents)

New in Grade 7 — teach yourself first

In Grade 6 you solved one-step equations and found percents of numbers. In Grade 7 you will solve *two-step* equations and use percents to find **markup, discount, tax, tip, and percent increase or decrease**. Read both examples, then try.

VOCABULARY

Two-step equation

An equation that takes two inverse operations to solve, like $2x + 3 = 11$.

Inverse operations

Operations that undo each other (+ and −, × and ÷).

Markup

An amount added to a price; the price goes up.

Discount

An amount taken off a price; the price goes down.

Percent change

$(\text{amount of change} \div \text{original}) \times 100\%$.

Final amount

The price after tax, tip, markup, or discount is applied.

WATCH ONE FIRST — TWO-STEP EQUATIONS

Example: Solve $2x + 3 = 11$.

Undo $+3$ first: subtract 3 from both sides $\rightarrow 2x = 8$.

Undo $\times 2$: divide both sides by 2 $\rightarrow x = 4$.

Check: $2(4) + 3 = 8 + 3 = 11$. ✓

WATCH ONE MORE — PERCENT CHANGE

Example 1 (discount): A \$40 shirt is 25% off.

Discount = $0.25 \times 40 = 10$. Sale price = $40 - 10 = \$30$.

Example 2 (percent increase): A price goes from \$20 to \$25. What is the percent increase?

Change = $25 - 20 = 5$. Percent change = $5 \div 20 = 0.25 = 25\%$ increase.

Set A • Solve two-step equations Undo $+$ or $-$ first, then $\times$ or $\div$.

1. $2x + 5 = 13$

5. $4x - 6 = 10$

2. $3x - 4 = 11$

6. $6x + 1 = 19$

3. $5x + 2 = 27$

7. $x/3 - 2 = 4$

4. $x/2 + 3 = 7$

8. $10 = 2x + 4$

Set B • Markup, discount, tax, tip Find the change, then the final amount.

1. A \$50 jacket is 20% off. Find the discount and the sale price.

2. A \$30 meal has a 15% tip. Find the tip and the total.

3. A \$25 item has 8% sales tax. Find the tax and the total.

4. A store buys a toy for \$12 and marks it up 50%. Find the markup and the selling price.

5. A \$80 pair of shoes is 25% off, then 6% tax is added on the sale price. Find the final price.

Set C • Percent increase & decrease change $\div$ original.

1. A price goes from \$40 to \$50. Percent increase?

2. A price goes from \$80 to \$60. Percent decrease?

3. A plant grows from 20 cm to 26 cm. Percent increase?
4. A phone drops from \$300 to \$240. Percent decrease?
5. A salary goes from \$1000 to \$1100. Percent increase?

Level 1 · Scaffolded

LEVEL 1

1. $2x + 1 = 9$
2. $3x - 2 = 7$
3. A \$20 shirt is 10% off. Discount?
4. A price goes from \$10 to \$12. Percent increase?

Level 2 · Stretch

LEVEL 2

1. A \$120 bike is 30% off, then 5% tax is added on the sale price. Find the final price.
2. A store raises a \$50 price by 20%, then later discounts the new price by 20%. Is the final price back to \$50? Find it and explain.

QUICK CHECK

1. Solve $4x + 3 = 23$.
2. A \$60 item is 25% off. Sale price?
3. A \$40 bill with a 20% tip — total?
4. A price goes from \$25 to \$30. Percent increase?

Cumulative / Spiral Review

These two mixed sets pull from every Part — mostly Grade 6 with a little Grade 7 preview mixed in. Use them right before school starts to make sure everything sticks.

Spiral Set A Mixed review.

- | | |
|---|--|
| 1. Fill in: $3 : 5 = 12 : \underline{\hspace{1cm}}$ | 8. Solve $x/4 = 6$. |
| 2. 180 miles in 3 hours — miles per hour? | 9. Find $ -7 $. |
| 3. $5/6 - 1/2$ | 10. Which quadrant is $(-3, 4)$ in? |
| 4. $2\frac{1}{2} \div \frac{1}{4}$ | 11. Triangle base 8, height 5 — area? |
| 5. 30% of 90 | 12. Data: 4, 6, 8, 10 — mean? |
| 6. 15 is 25% of what number? | 13. $-5 + (-8)$ (Grade 7 preview) |
| 7. Evaluate $3x + 4$ when $x = 5$. | 14. For $y = 4x$, find y when $x = 6$ (Grade 7 preview) |

Spiral Set B Mixed review.

- | | |
|--|--|
| 1. Better buy: 12 oz for \$3.00 or 20 oz for \$4.60? | 8. Order least to greatest: $-3, 1, -6, 0$. |
| 2. $3/4 + 1/6$ | 9. Distance between $(2, -1)$ and $(2, 6)$. |
| 3. $4.8 \div 0.6$ | 10. Cube edge 3 — volume? |
| 4. Write 0.45 as a percent. | 11. Data: 5, 7, 7, 9 — median? |
| 5. A \$40 game is 15% off — sale price? | 12. $3 - (-5)$ (Grade 7 preview) |
| 6. Distribute: $4(x + 5)$. | 13. Solve $2x + 3 = 15$ (Grade 7 preview) |
| 7. Solve $x + 9 = 17$. | 14. A price goes from \$20 to \$25 — percent increase? (Grade 7 preview) |

Answer Keys

For teachers/parents. Check student work against these. Some problems allow equivalent forms (e.g., $9/8 = 1 \frac{1}{8}$).

Part 1 — Ratios, Rates & Unit Rate

Set A:

- | | | |
|---|-----------|----------------------------------|
| 1. 1 : 3 (A is the vowel; M, T, H are consonants) | 3. 8 : 13 | 6. 21 |
| 2. 5 : 8 | 4. 20 | 7. No; $3:4 = 9:12$, not $9:16$ |
| | 5. 2 | 8. e.g. 10:4 and 15:6 |

Set B:

- | | | |
|---------------------------|------------------|-----------|
| 1. (4,6),(6,9),(8,12) | 4. 3 cups sugar | 7. 9 dogs |
| 2. (10,8),(15,12),(20,16) | 5. 25 cups white | |
| 3. 24 apples | 6. 15 girls | |

Set C:

- | | | |
|---------------------|-------------------|---|
| 1. 50 mph | 4. \$9 per ticket | 6. 30 oz: \$0.18/oz vs 20 oz: \$0.20/oz — the 30 oz is the better buy |
| 2. \$0.50 per apple | 5. 6.25 min/km | |
| 3. 15 pages/min | | |

Set D:

- | | |
|---|---|
| 1. Ratios scale by multiplying, not adding; $3:4 = 6:8$ | 2. Unit rate = $100 \div 4 = 25$ miles per hour |
|---|---|

Level 1:

- | | |
|----------|----------------|
| 1. 1 : 3 | 3. \$0.50 |
| 2. 12 | 4. (2,4),(3,6) |

Level 2:

- | | |
|---|---|
| 1. A: $2/3 \approx 0.667$ fruit per juice;
B: $3/5 = 0.6$ — Smoothie A is fruitier | 2. 12 nickels and 8 dimes. (3:2 means each "group" has 3 nickels and 2 dimes, worth $3 \times \$0.05 + 2 \times \$0.10 = \$0.15 + \$0.20 = \$0.35$; $\$1.40 \div \$0.35 = 4$ groups $\rightarrow 3 \times 4 = 12$ nickels = $\$0.60$ and $2 \times 4 = 8$ dimes = $\$0.80$; total $\$1.40$.) |
|---|---|

Quick Check:

- | | | |
|-----------|------------|---|
| 1. 12 | 3. 12 boys | 4. 24 oz: \$0.18/oz vs 16 oz: \$0.20/oz — 24 oz is better |
| 2. 60 mph | | |

Part 2 — Fraction & Decimal Operations

Set A:

- | | | |
|--------------------|-------------------|-------------------|
| 1. $\frac{5}{6}$ | 4. $\frac{3}{8}$ | 7. $2\frac{1}{3}$ |
| 2. $\frac{11}{12}$ | 5. $3\frac{3}{4}$ | |
| 3. $\frac{1}{2}$ | 6. $2\frac{1}{2}$ | |

Set B:

- | | | |
|------------------|------------------|------|
| 1. $\frac{2}{5}$ | 3. $\frac{1}{3}$ | 5. 3 |
| 2. $\frac{1}{6}$ | 4. 1 | 6. 4 |

Set C:

- | | | |
|-------------------|-------------------|---------------|
| 1. 2 | 4. $2\frac{1}{2}$ | 7. 9 |
| 2. $\frac{9}{10}$ | 5. 5 | 8. 8 servings |
| 3. $\frac{2}{3}$ | 6. 3 | |

Set D:

- | | | |
|----------|---------|-------|
| 1. 16.15 | 3. 2.24 | 5. 32 |
| 2. 13.62 | 4. 0.15 | 6. 12 |

Set E:

- | | | |
|--------------|-----------|---|
| 1. 10 pieces | 2. 2 cups | 3. They forgot to flip and multiply;
$\frac{1}{2} \div \frac{1}{4} = \frac{1}{2} \times \frac{4}{1} = 2$ |
|--------------|-----------|---|

Level 1:

- | | |
|------------------|------|
| 1. $\frac{1}{2}$ | 3. 2 |
| 2. $\frac{1}{4}$ | 4. 4 |

Level 2:

- | | |
|--|--|
| 1. $(\frac{8}{12} + \frac{3}{12}) = \frac{11}{12}$; $\frac{11}{12} \div \frac{5}{6} = \frac{11}{12} \times \frac{6}{5} = \frac{66}{60} = \frac{11}{10} = 1\frac{1}{10}$ | 2. $\frac{1}{3}$ of $4\frac{1}{2}$ left = $1\frac{1}{2}$ miles |
|--|--|

Quick Check:

- | | |
|-------------------|-------------------|
| 1. $\frac{7}{12}$ | 3. $3\frac{1}{3}$ |
| 2. $\frac{2}{3}$ | 4. 8 |

Part 3 — Percents & Proportional Applications

Set A:

- | | | |
|------------------|------------------|-----------|
| 1. $0.5 = 50\%$ | 4. $0.45 = 9/20$ | 7. 70% |
| 2. $0.75 = 75\%$ | 5. 20% | 8. $1/8$ |
| 3. $3/5 = 60\%$ | 6. 8% | |

Set B:

- | | | |
|-------|-------|-------|
| 1. 10 | 4. 10 | 7. 12 |
| 2. 9 | 5. 21 | 8. 20 |
| 3. 30 | 6. 30 | |

Set C:

- | | | |
|-------|-------|--------|
| 1. 48 | 3. 90 | 5. 200 |
| 2. 60 | 4. 24 | 6. 140 |

Set D:

- | | | |
|------------------------------|----------------|-----------|
| 1. \$15 off; sale price \$45 | 3. 12 students | 5. \$2.30 |
| 2. 75% | 4. \$7 | |

Level 1:

- | | |
|-------|--------|
| 1. 10 | 3. 25% |
| 2. 6 | 4. 10 |

Level 2:

- | | |
|--|--|
| 1. Sale = $200 - 60 = \$140$; tax = $0.06 \times 140 = \$8.40$; final = \$148.40 | 2. Whole = $32 \div 0.8 = 40$; 40 is 40% of 100 |
|--|--|

Quick Check:

- | | |
|--------|---------|
| 1. 9 | 3. 30 |
| 2. 35% | 4. \$40 |

Part 4 — Expressions, Equations & Inequalities

Set A:

- | | | |
|------------|-------------|------------------------------|
| 1. $n + 7$ | 3. $y/5$ | 5. coefficient 9, constant 2 |
| 2. $4x$ | 4. $2m - 6$ | 6. 3 terms |

Set B:

- | | | |
|-------|-------|-------|
| 1. 13 | 3. 10 | 5. 5 |
| 2. 23 | 4. 30 | 6. 16 |

Set C:

- | | | |
|--------------|-------------|--|
| 1. $3x + 12$ | 3. $7x$ | 5. $3(2x + 3)$ |
| 2. $10a + 5$ | 4. $8y + 2$ | 6. Yes; distributing $2(x+3)$ gives $2x + 6$ |

Set D:

- | | | |
|-------------|-------------|--------------|
| 1. $x = 7$ | 3. $x = 7$ | 5. $x = 6.5$ |
| 2. $x = 14$ | 4. $x = 21$ | 6. $x = 4$ |

Set E:

- | | | |
|----------------|---------------|--|
| 1. $x \geq 10$ | 3. $x > 7$ | 5. $x \leq 5$; e.g. $x = 0$ and $x = 5$ |
| 2. $a < 13$ | 4. $x \leq 7$ | |

Level 1:

- | | |
|------------|------------|
| 1. 8 | 3. $x = 4$ |
| 2. $x = 7$ | 4. $n + 5$ |

Level 2:

- | | |
|--|---|
| 1. $3(x+4)+2x = 5x + 12$; at $x=2 \rightarrow 22$ | 2. $50 + 8w$; after 6 weeks = $50 + 48 = \$98$ |
|--|---|

Quick Check:

- | | |
|-------------|---------------|
| 1. 19 | 3. $4x + 12$ |
| 2. $x = 20$ | 4. $x \geq 4$ |

Part 5 — Rational Numbers & the Coordinate Plane

Set A:

- | | | |
|-------|---|------------------------|
| 1. -7 | 4. 15 | 7. 3°F |
| 2. 12 | 5. -3 | 8. -12 |
| 3. 9 | 6. $ -6 = 6$ is greater than $ 4 = 4$ | |

Set B:

- | | | |
|-----------------|----------------------|----------------------------------|
| 1. -7, -2, 1, 5 | 3. $-1/2, 0, 1/2, 1$ | 5. $-3 > -6$ |
| 2. -4, -1, 0, 3 | 4. -2, -1.5, -1, 0 | 6. $2/3 > 0.6$ ($0.667 > 0.6$) |

Set C:

- | | | |
|-----------------|------------------|------------|
| 1. Quadrant IV | 3. Quadrant II | 5. (3, -4) |
| 2. Quadrant III | 4. On the y-axis | 6. (2, 5) |

Set D:

- | | | |
|------|------|---|
| 1. 7 | 3. 7 | 4. $ -8 = 8$; absolute value is distance from 0, never negative |
| 2. 8 | | |

Level 1:

- | | |
|-------|---------------|
| 1. -4 | 3. 1 |
| 2. 3 | 4. Quadrant I |

Level 2:

- | | |
|----------------------------|-------------------------------|
| 1. $-30 + 12 - 7 = -25$ ft | 2. length 7, width 3, area 21 |
|----------------------------|-------------------------------|

Quick Check:

- | | |
|-----------------|-----------------|
| 1. 11 | 3. Quadrant III |
| 2. -9, -4, 0, 2 | 4. 8 |

Part 6 — Geometry: Area, Surface Area & Volume

Set A:

1. 40

2. 42

3. 36

4. 15

5. 32

6. 88

Set B:

1. $60 - 12 = 48$

2. $32 + 12 = 44$

3. $60 + 25 = 85$

Set C:

1. 54 cm^2

2. 52

3. 150 in^2

4. cube; 24

Set D:

1. 60

2. 64

3. 3

4. 20

5. $\frac{1}{8}$

6. 8

Level 1:

1. 24

2. 16

3. 8

4. 24

Level 2:

1. 2400 in^3 ; $2400 \div 231 \approx 10$
gallons

2. one trapezoid = $\frac{1}{2}(8+14)(6) =$
66; combined = 132

Quick Check:

1. 45

2. 30

3. 54

4. 96

Part 7 — Statistics & Data

Set A:

1. No; it has one answer
2. Yes; answers vary
3. e.g. "How many pets does each student have?"
4. No; one answer for one day

Set B:

1. 7
2. 7
3. 7 and 9 (bimodal)
4. 6
5. 13
6. 8
7. mode 4; range 8
8. mean 9; median 9

Set C:

1. mean 5; $MAD = (3+1+1+3)/4 = 2$
2. mean 11; $MAD = (1+1+1+1)/4 = 1$
3. Set Y (range 12)

Set D:

1. 7 points; mode 4
2. 25 students; 80–89 has the most
3. range 14; IQR 6
4. skewed right

Level 1:

1. 4
2. 5
3. 4
4. 7

Level 2:

1. fifth number = 12
2. mean 6; $MAD = (3+1+1+1+4)/5 = 2$

Quick Check:

1. 7
2. 6
3. 7
4. mean 4; $MAD = (3+1+1+3)/4 = 2$

Part 8 — PREVIEW: Integer Operations

Set A:

1. -10

2. -5

3. 4

4. -9

5. 0

6. -3

7. 7

8. -6

Set B:

1. -5

2. -7

3. 10

4. -4

5. -9

6. 0

7. 14

8. -12

Set C:

1. -15

2. 24

3. -14

4. -4

5. 6

6. 0

7. -4

8. -8

Set D:

1. -10°F

2. -75 ft

3. $-\$6$ (still owes $\$6$)

4. $-5 - 3 = -8$; subtracting 3 means adding -3

Level 1:

1. -5

2. 4

3. -3

4. -6

Level 2:

1. $-4 + 7 + 2 - 9 = -4$

2. $-12 + 4 = -8$

Quick Check:

1. 4

2. 5

3. 21

4. -7

Part 9 — PREVIEW: Proportional Relationships

Set A:

- | | | |
|--------------|--------------|-----------------------------|
| 1. $k = 4$ | 3. $k = 4$ | 5. $k = 0.5$ (\$0.50/apple) |
| 2. $k = 2.5$ | 4. $k = 4.5$ | 6. $k = 30$ (mpg) |

Set B:

- | | | |
|-------------|-------------|------------|
| 1. $y = 5x$ | 3. 15 | 5. 60 |
| 2. 28 | 4. $y = 6x$ | 6. $x = 7$ |

Set C:

- | | | |
|-------------------------------|-----------------|---------------------------------------|
| 1. Yes; $k = 3$ | 3. Yes; $k = 5$ | 4. No; it does not pass through (0,0) |
| 2. No; 3, 2.5, 2.33 not equal | | |

Set D:

- | | |
|-------------------------|-------------------------|
| 1. $y = 8x$; 120 pages | 2. $y = 0.40x$; \$4.80 |
|-------------------------|-------------------------|

Level 1:

- | | |
|------------|-------------|
| 1. $k = 4$ | 3. $y = 2x$ |
| 2. 15 | 4. $k = 7$ |

Level 2:

- | | |
|---|--|
| 1. $y = 7x$ gives 35 vs $y = 4x$ gives 20; larger by 15 | 2. $y = 4x$ (muffins per cup); $40 \div 4 = 10$ cups |
|---|--|

Quick Check:

- | | |
|------------|---|
| 1. $k = 7$ | 3. No ($6/2=3$, $12/4=3$, $14/5=2.8$) |
| 2. 54 | 4. $y = 9x$ |

Part 10 — PREVIEW: Two-Step Equations & Percent Change

Set A:

- | | | |
|------------|------------|-------------|
| 1. $x = 4$ | 4. $x = 8$ | 7. $x = 18$ |
| 2. $x = 5$ | 5. $x = 4$ | 8. $x = 3$ |
| 3. $x = 5$ | 6. $x = 3$ | |

Set B:

- | | | |
|------------------------------|-----------------------------------|---|
| 1. discount \$10; sale \$40 | 3. tax \$2; total \$27 | 5. sale = $80 - 20 = \$60$; tax |
| 2. tip \$4.50; total \$34.50 | 4. markup \$6; selling price \$18 | $0.06 \times 60 = \$3.60$; final \$63.60 |

Set C:

- | | | |
|-----------------|-----------------|-----------------|
| 1. 25% increase | 3. 30% increase | 5. 10% increase |
| 2. 25% decrease | 4. 20% decrease | |

Level 1:

- | | |
|------------|-----------------|
| 1. $x = 4$ | 3. \$2 off |
| 2. $x = 3$ | 4. 20% increase |

Level 2:

- | | |
|--|--|
| 1. sale = $120 - 36 = \$84$; tax
$0.05 \times 84 = \$4.20$; final \$88.20 | 2. $50 \rightarrow +20\% = \$60 \rightarrow -20\% =$
$60 - 12 = \$48$; no, it is \$48, not
\$50 (20% of 60 is larger than
20% of 50) |
|--|--|

Quick Check:

- | | |
|------------|-----------------|
| 1. $x = 5$ | 3. \$48 |
| 2. \$45 | 4. 20% increase |

Spiral Set A

- | | | |
|------------------|-----------------|---------|
| 1. 20 | 6. 60 | 11. 20 |
| 2. 60 mph | 7. 19 | 12. 7 |
| 3. $\frac{1}{3}$ | 8. $x = 24$ | 13. -13 |
| 4. 10 | 9. 7 | 14. 24 |
| 5. 27 | 10. Quadrant II | |

Spiral Set B

- | | | |
|---|-----------------|------------------|
| 1. 20 oz: \$0.23/oz vs 12 oz:
\$0.25/oz — 20 oz better | 5. \$34 | 10. 27 |
| 2. $\frac{11}{12}$ | 6. $4x + 20$ | 11. 7 |
| 3. 8 | 7. $x = 8$ | 12. 8 |
| 4. 45% | 8. -6, -3, 0, 1 | 13. $x = 6$ |
| | 9. 7 | 14. 25% increase |